Title: Complications associated with percutaneous native kidney biopsies in adults in low- and middle-income countries: A Systematic review and meta-analysis

SUPPLEMENTARY MATERIALS

Supplementary information is available at KI Report's website

PRSIMA checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title (Page 1)
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract (Page 2)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction (Page 3)
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction (Page 4)
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Page 4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Page 5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Page 5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 5-6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 6-7

Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Page 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Page 6-7

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 6 -7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Page 6 -7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1 (pg22)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Table S2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Results (pg 7-9)
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results (pg 7-9)
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Discussion (Pg 9– 14)
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results (pg 7–9)
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion (Pg 9–13)
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion (Pg 13)
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion (Pg 9– 13)
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 15

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

Table S1: MEDLINE search strategy

1	Kidney biopsy OR Renal biopsy
2	Complications OR Adverse events OR Adverse effects OR haematoma OR Hematoma OR hemorrhage OR haemorrhage OR bleeding OR blood transfusion OR Red blood cell transfusion OR Erythrocyte transfusion OR red cell transfusion OR cauterisation OR embolisation OR embolization OR Bladder Obstruction OR Arteriovenous fistula OR Arteriovenous aneurysm OR pseudo aneurysm OR Infection OR Angiography OR Needle OR Fine needle OR Ultrasound guided OR nephrectomy OR death
3	((Afghanistan [tiab] OR Albania [tiab] OR Algeria [tiab] OR American Samoa [tiab] OR Angola [tiab] OR Argentina[tiab] OR Armenia [tiab] OR Azerbaijan [tiab] OR Bangladesh [tiab] OR Belarus [tiab] OR Belize [tiab] OR Benin [tiab] OR Bhutan [tiab] OR Bolivia [tiab] OR Bosnia and Herzegovina [tiab] OR Botswana [tiab] OR Brazil [tiab] OR Bulgaria [tiab] OR Burkina Faso [tiab] OR Burundi [tiab] OR Cabo Verde [tiab] OR Cambodia [tiab] OR Cameroon [tiab] OR Central African Republic [tiab] OR Chad [tiab] OR China [tiab] OR Colombia [tiab] OR Comoros [tiab] OR Congo, Dem. Rep. [tiab] OR Congo, Rep. [tiab] OR Costa Rica [tiab] OR Ivory Coast [tiab] OR Cote d'Ivoire [tiab] OR Cuba [tiab] OR Djibouti [tiab] OR Dominica [tiab] OR Dominican Republic [tiab] OR Ecuador [tiab] OR Egypt.[tiab] OR El Salvador [tiab] OR Equatorial Guinea [tiab] OR Eritrea [tiab] OR Ethiopia [tiab] OR Fiji [tiab] OR Gabon [tiab] OR Gambia [tiab] OR Georgia [tiab] OR Ghana [tiab] OR Grenada [tiab] OR Guatemala [tiab] OR Guinea [tiab] OR Guinea-Bissau [tiab] OR Guyana [tiab] OR Haiti [tiab] OR Honduras [tiab] OR India [tiab] OR Indonesia [tiab] OR Iran, Islamic Rep. [tiab] OR Iraq [tiab] OR Jamaica [tiab] OR Jordan [tiab] OR Kazakhstan [tiab] OR Kenya [tiab] OR Kiribati [tiab] OR Korea, Dem. People's Rep. [tiab] OR Kosovo [tiab] OR Kyrgyz Republic [tiab] OR Lao PDR [tiab] OR Lebanon [tiab] OR Lesotho [tiab] OR Liberia [tiab] OR Libya [tiab] OR Macedonia [tiab] OR Madagascar [tiab] OR Malawi [tiab] OR Malaysia [tiab] OR Maldives [tiab] OR Mali [tiab] OR Marshall Islands [tiab] OR Mauritania [tiab] OR Mauritius [tiab] OR Mexico [tiab] OR Micronesia [tiab] OR Moldova [tiab] OR Mongolia [tiab] OR Montenegro [tiab] OR Morocco [tiab] OR Mozambique [tiab] OR Myanmar [tiab] OR Namibia [tiab] OR Nepal [tiab] OR Nicaragua [tiab] OR Niger [tiab] OR Nigeria [tiab] OR Pakistan [tiab] OR Palau [tiab] OR Panama [tiab] OR Papua New Guinea [tiab] OR Paraguay [tiab] OR Peru [tiab] OR Philippines [tiab] OR Romania [tiab] OR Russian Federation [tiab] OR Rwanda [tiab] OR Samoa [tiab] OR São Tomé and Princip [tiab] OR Senegal [tiab] OR Serbia [tiab] OR Sierra Leone [tiab] OR Solomon Islands [tiab] OR Somalia [tiab] OR South Africa [tiab] OR South Sudan [tiab] OR Sri Lanka [tiab] OR St. Lucia [tiab] OR St. Vincent and the Grenadines [tiab] OR Sudan [tiab] OR Suriname [tiab] OR Swaziland [tiab] OR Syrian Arab Republic [tiab] OR Tajikistan [tiab] OR Tanzania [tiab] OR Thailand [tiab] OR Timor-Leste [tiab] OR Togo [tiab] OR Tonga [tiab] OR Tunisia [tiab] OR Turkey [tiab] OR Turkmenistan [tiab] OR Tuvalu [tiab] OR Uganda [tiab] OR Ukraine [tiab] OR Uzbekistan [tiab] OR Vanuatu [tiab] OR Venezuela [tiab] OR Vietnam [tiab] OR West Bank and Gaza [tiab] OR Yemen, Rep. [tiab] OR Zambia [tiab] OR Zimbabwe [tiab]))
4	1 + 2 + 3

Table S2: Adapted tool for risk of bias assessment

Quality assessment checklist for the systematic review and Meta-analysis
"The incidence of major complications after percutaneous native renal biopsies in adults from low to middle income countries
Name of author(s):……
Year of publication: ……..
Study title: Kidney Biopsy:……..

Risk of bias items	Risk of bias levels	Points scored
1. Was the study's target population close to all patients with indications for Native renal biopsies at the study centre, e.g. Not biopsies of specific target groups like elderly, diabetics, transplants etc?	**Yes** (**LOW RISK**): The study's target population was a close representation of people going for renal biopsies in that centre	0
	No (**HIGH RISK**): The study's target population was clearly NOT representative of the renal biopsy population	1
2. Where people selected for biopsies a true reflection of those with indications for renal biopsies at the centre?	**Yes** (**LOW RISK**): The sampling frame was a true or close representation of the target population.	0
	No (**HIGH RISK**): The sampling frame was NOT a true or close representation of the target population.	1
3. Where all patients needing bx done or it was selective?	**Yes** (**LOW RISK**): The study reported on all patients biopsied at that centre during the study period	0
	No (**HIGH RISK**): The study did NOT report on all patients biopsied at that centre during the study period and was selective	1
4. Where all patients biopsied followed up to record all complications?	**Yes** (**LOW RISK**): The response rate for the study was ≥75%, OR, an analysis was performed that showed no significant difference in relevant demographic characteristics between responders and non- responders	0
	No (**HIGH RISK**): The response rate was <75%, and if any analysis comparing responders and non-responders was done, it showed a significant difference in relevant demographic characteristics between responders and non-responders	1
5. Were data collected directly from the subjects (as opposed to a proxy)?	**Yes** (**LOW RISK**): All data were collected directly from the subjects.	0
	No (**HIGH RISK**): In some instances, data were collected from a proxy.	1
6. Was an acceptable case definition used in the study? Where major or minor complications defined appropriately	**Yes** (**LOW RISK**): An acceptable case definition was used.	0
	No (**HIGH RISK**): An acceptable case definition was NOT used	1
7. Was the study instrument that measured the parameter of interest (e.g. Prevalence of haematuria, Pain, haematomas) shown to have reliability and validity (if necessary)?	**Yes** (**LOW RISK**): The study instrument had been shown to have reliability and validity (if this was necessary), e.g. test-re- test, piloting, validation in a previous study, etc.	0
	No (**HIGH RISK**): The study instrument had NOT been shown to have reliability or validity (if this was necessary).	1
8. Was the same mode of data collection used for all subjects?	**Yes** (**LOW RISK**): The same mode of data collection was used for all subjects **No** (**HIGH RISK**): The same mode of data collection was NOT used for all subjects	0
9. Were the numerator(s) and denominator(s) for the parameter of interest appropriate	**Yes** (**LOW RISK**): The paper presented appropriate numerator(s) AND denominator(s) for the parameter of interest (e.g. the prevalence of low back pain).	0
	No (**HIGH RISK**): The paper did present numerator(s) AND denominator(s) for the parameter of interest but one or more of these were inappropriate.	1
10. Summary on the overall risk of study bias	**LOW RISK**	0-3
	MODERATE RISK	4-6
	HIGH RISK SCORE 7	7-9

Table S3: Complication rates following percutaneous native kidney biopsies

Author [REF]	Country	Income group	Average glomeruli (n)	Adequacy of biopsy (%)	Major Macroscopic haematuria (%)	Nephrectomy (%)	Major haematoma (%)	Red Cell transfusion (%)	Angiographic intervention (%)	Infection (%)	Death (%)
Tuladhar et al[54]	Nepal	LIC	21.8	NR	0	0	0	0	0	0	0
Ghimire et al[20]	Nepal	LIC	NR	NR	0	0	0	0	0	0	0
Manandhar et al[21]	Nepal	LIC	23.4	NR	0	0	0	0	0	0	0
Pokhrel et al[15]	Nepal	LIC	23.5±8.3	94.6	29.7	0	13.5	NR	NR	NR	0
Pokhrel 2 et al[15]	Nepal	LIC	23.9±10.0	92.1	39.5	0	7.9	NR	NR	NR	0
Abdou et al[22]	Senegal	LIC	NR	NR	0	0	0	0	0	0	0
Aatif et al[23]	Morocco	LMIC	15.8	NR	0	0.6	0	0	0	0	0
Zajjari et al[8]	Morocco	LMIC	NR	NR	0	0.8	0	0	0	0	0
Sobh et al[24]	Egypt	LMIC	NR	71.0	6.7	0	0	5.1	0	10	0
Hachicha et al[25]	Tunisia	LMIC	NR	73.3	0	0	1.7	1.7	0	0	0
Hachicha et al 2[25]	Tunisia	LMIC	NR	96.7	0	0	0	0	0	0	0
Nadium et al[26]	Sudan	LMIC	NR	NR	9.6	0	0	2.4	0	0	0
Musa et al[27]	Sudan	LMIC	NR	100.0	1.6	0	0	1.6	0	0	0
Obineche et al[28]	Nigeria	LMIC	NR	87.6	0	0	0	0	0	0	0
Krishna A et al[53]	India	LMIC	NR	95	5.2	NR	NR	1.1	0	0	0
Prakash et al[29]	India	LMIC	NR	79.0	4.0	0	0	1	0	0.3	0
Prasad et al[30]	India	LMIC	15.4	84.0	2.9	0	1.1	0.6	0.5	0	0
Yesudas et al[31]	India	LMIC	14.7	86.5	NR	NR	1.4	NR	NR	0	0
Sakhuja et al[32]	India	LMIC	NR	89.0	0.7	0	0	0.7	0	0	0
Golay et al[33]	India	LMIC	28.8	100.0	0	0	0	0.7	0	0.3	0
Arora et al[34]	India	LMIC	8.6	98.0	0	0	0	0	0	0	0
Ahmed et al[35]	Pakistan	LMIC	7.8	NR	0	0	0	0	0	0	0
Azmat et al[36]	Pakistan	LMIC	NR	92.5	NR	0	0	6.4	0	0	0
Mansoor et al[37]	Pakistan	LMIC	NR	NR	0	0	0	0	0	0	0
Yaqub et al[38]	Pakistan	LMIC	NR	97.7	3.9	0	NR	3.9	0	0	0
Habas et al[39]	Libya	UMIC	9.0	98.3	0.8	0.8	0.8	0	0.8	0	0.8
Mishra et al[40]	Libya	UMIC	17.5	NR	1.2	1.2	0	0	0	0	0
Ghnaimat et al [41]	Jordan	UMIC	NR	NR	0.5	0	0	0.5	0	0	0
Chen et al[55]	China	UMIC	NR	NR	NR	NR	NR	NR	NR	NR	0
Hu et al[43]	China	UMIC	NR	NR	0	0	1.7	0.2	0.1	0	0
Tao et al[44]	China	UMIC	NR	NR	NR	NR	7.9	NR	NR	NR	0
Wang et al[45]	China	UMIC	NR	NR	NR	0	1.6	3.9	0.9	0	0
Xu et al[4]	China	UMIC	NR	94.9	0	0	0	0.2	0.2	0	0
Pongsittisak W[16]	Thailand	UMIC	12	50	5	0	2	7	0	0	0
Pongsittisak W 2[16]	Thailand	UMIC	11.8	75	10.6	0	4.8	7.7	0	0	0
Kanjanabuchi et al[53]	Thailand	UMIC	NR	NR	0	0	0	0.6	0	0	0
Covic et al[47]	Romania	UMIC	9.1	95.4	2.4	NR	0.9	NR	NR	NR	0
Trajceska L[52]	Macedonia	UMIC	NR	99.1	4.9	0	0.6	1.7	0	0	0
Kovacevic et al[48]	Serbia	UMIC	NR	NR	1	0	0.5	0	0	0.2	0
Munoz et al[49]	Mexico	UMIC	NR	97.5	0.3	0.2	1.4	1.8	0.2	NR	0
Gonzalez-Michaca et al[50]	Mexico	UMIC	NR	NR	2.1	0.3	1.5	2.6	0.1	0.3	0.1
Kruger et al[51]	South Africa	UMIC	NR	99.1	0	0	0	NR	0	0	0

LIC – Low-income country; LMIC – Lower middle-income country; UMIC – Upper middle-income country; NR – “not reported”

Figure S1: Cumulative analysis for overall complications by study era

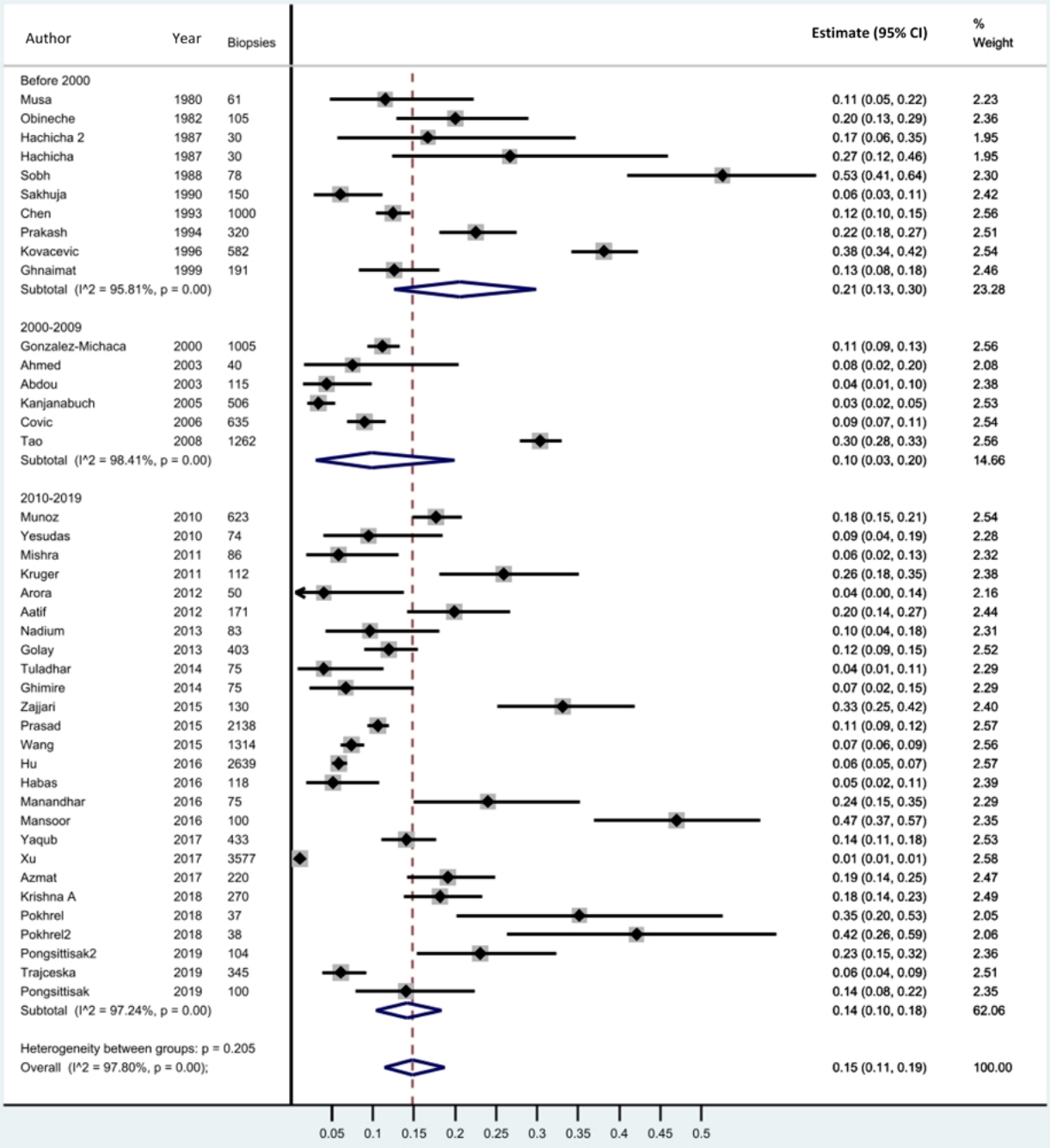


Figure S2: Overall complication rates grouped according to needle gauge

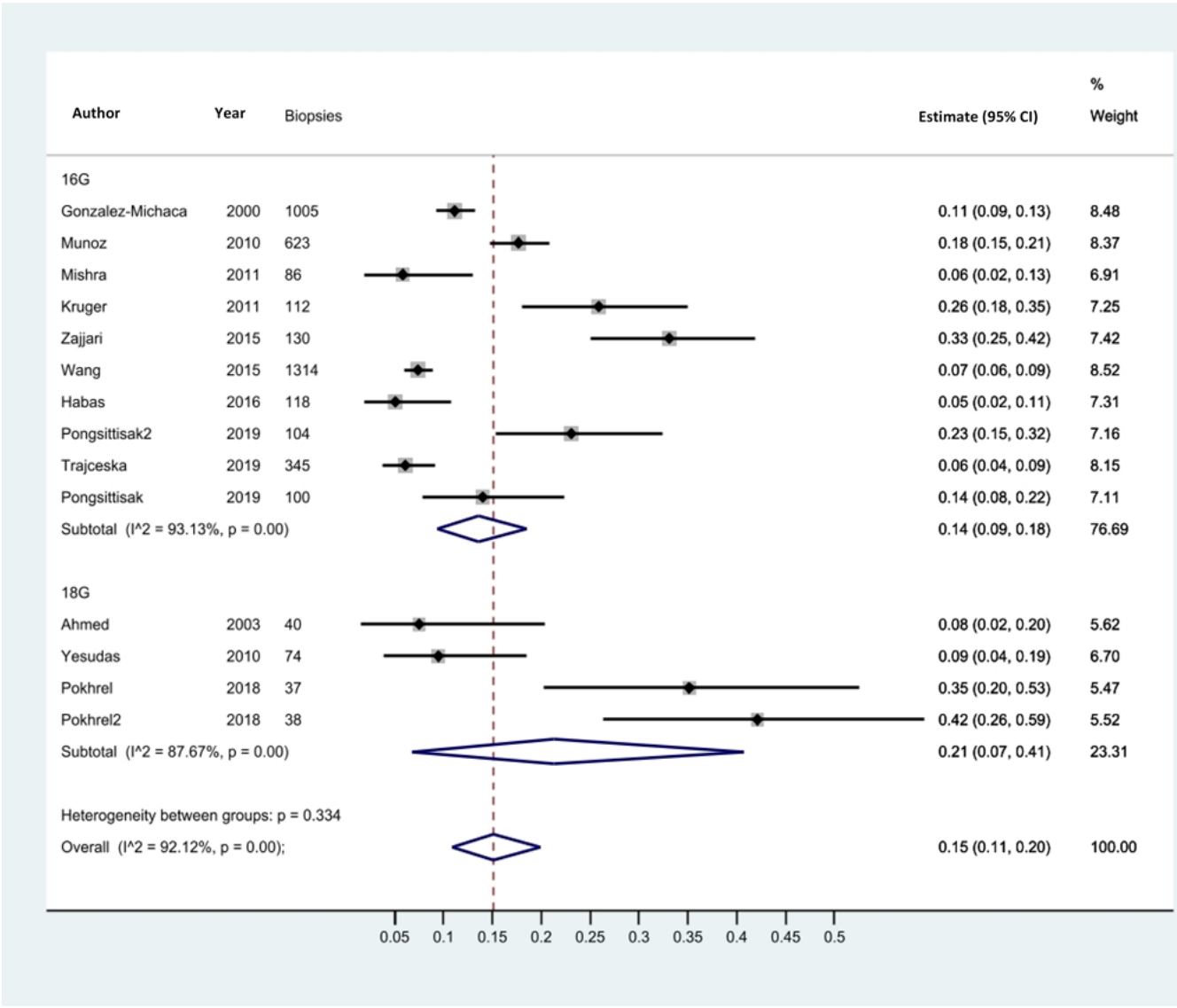


Figure S3: Overall complication rates grouped according to income group

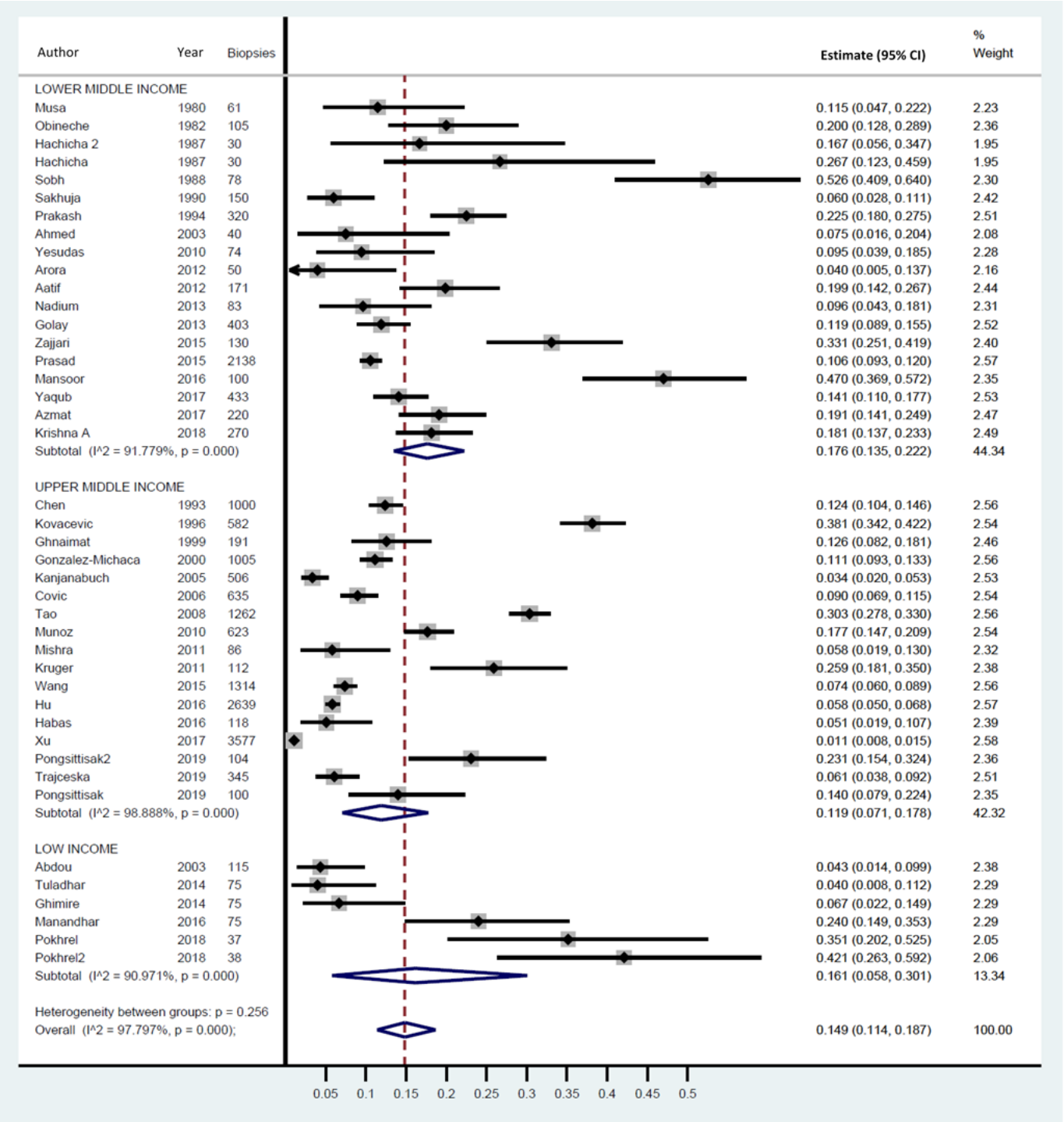


Figure S4: Overall complication rates according to region

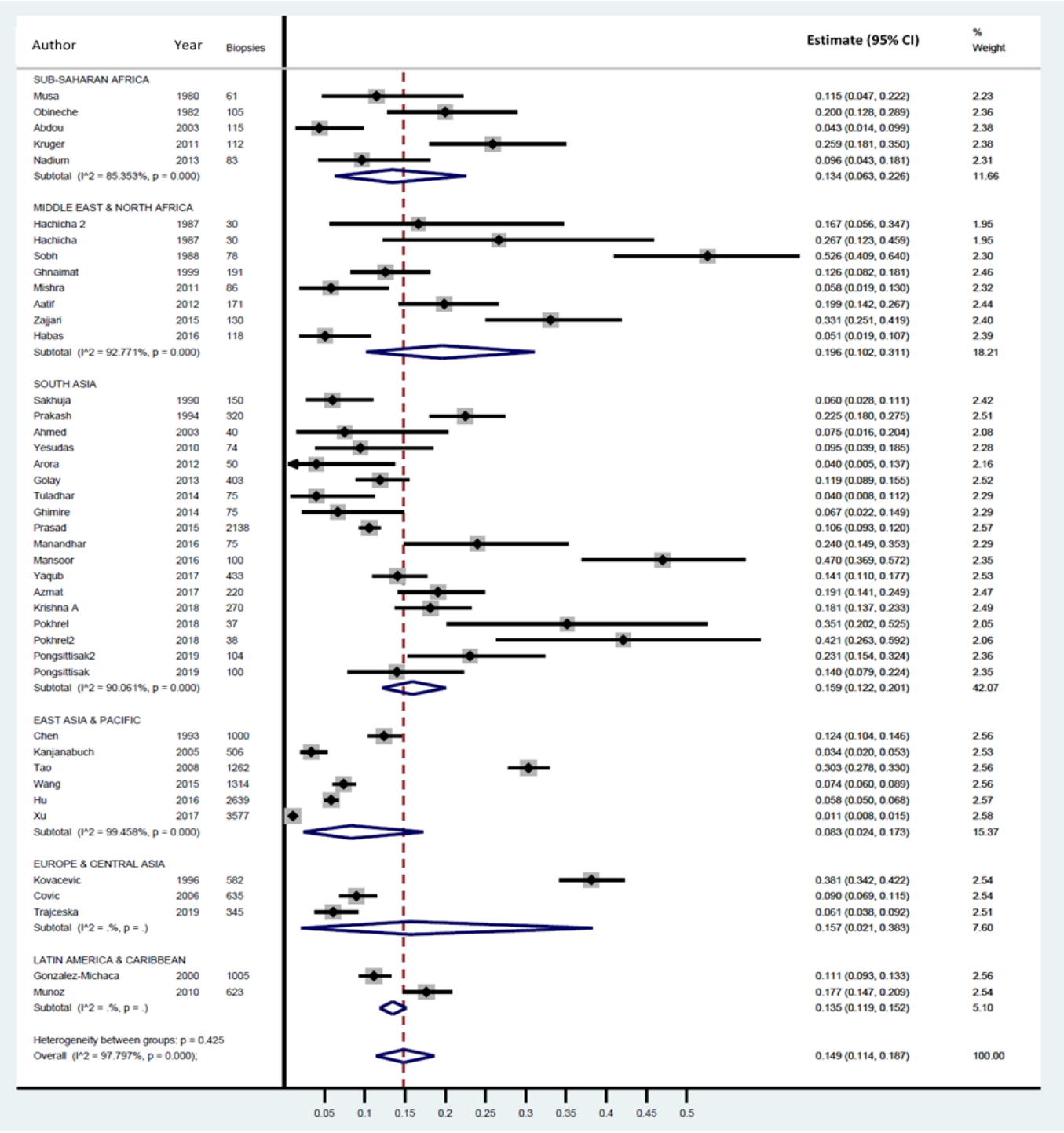


Figure S5: Overall complication rates grouped according to needle type used

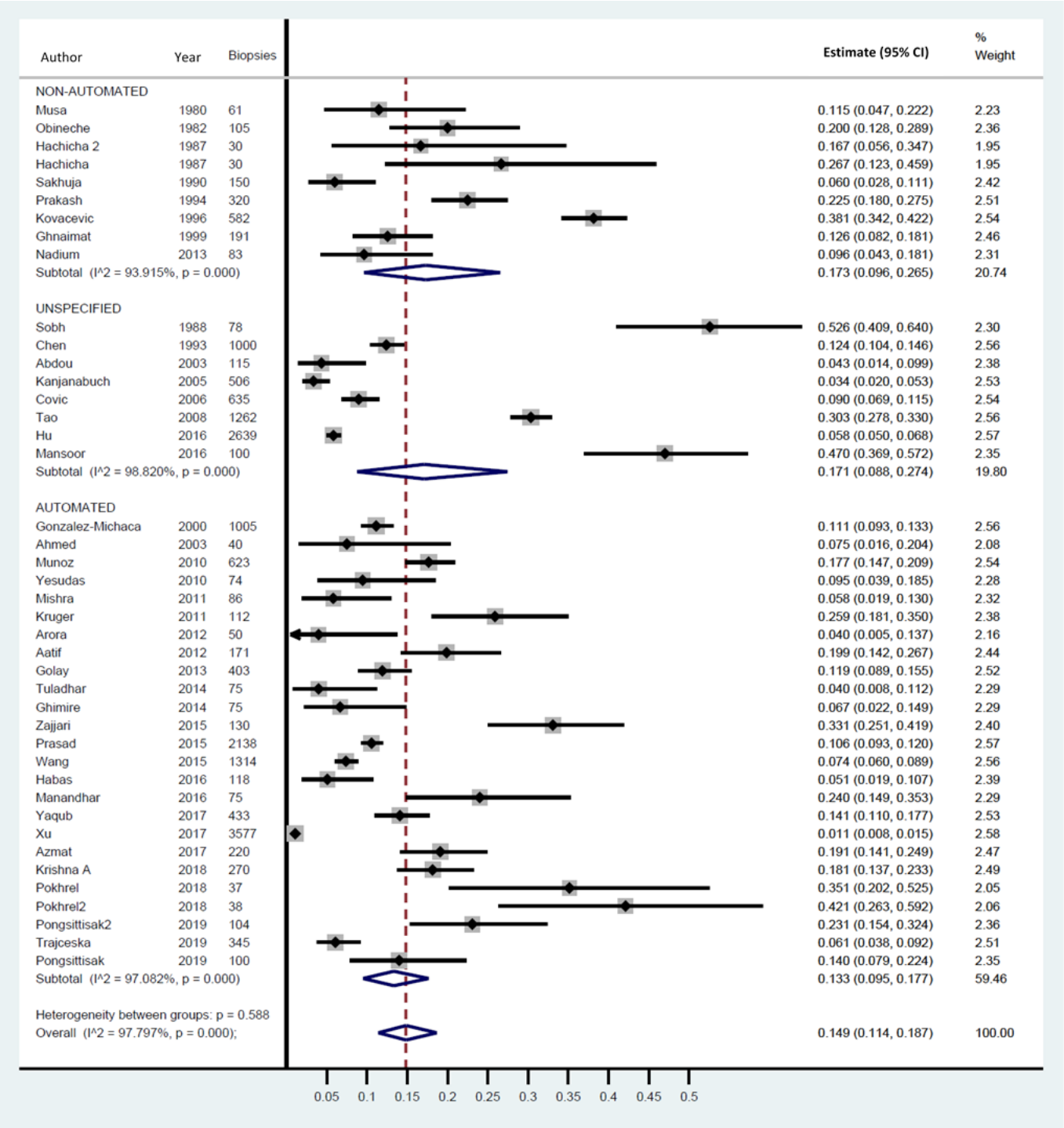


Figure S6: Major complications by needle type

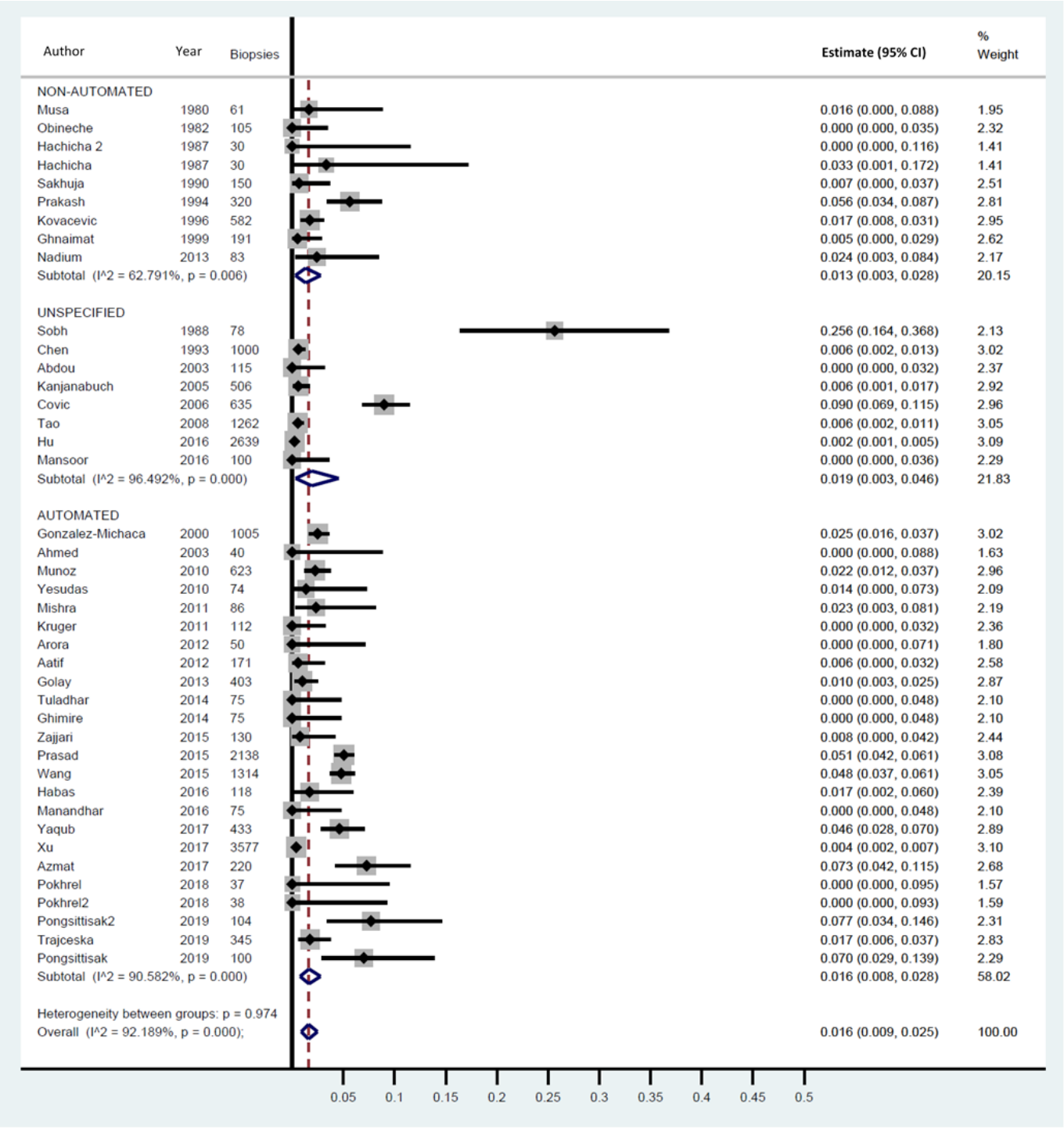


Figure S7: Major complications by needle gauge

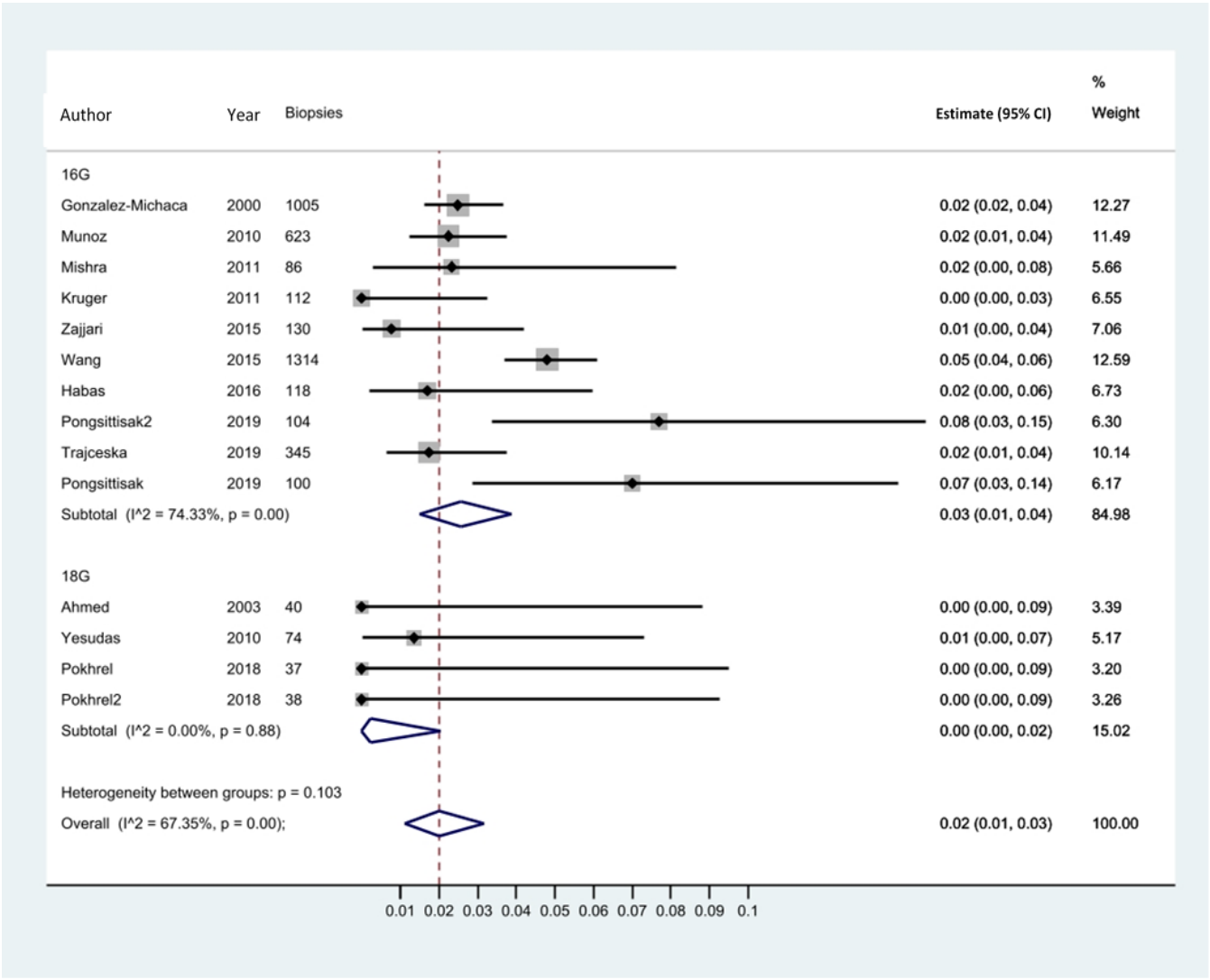


Figure S8: Cumulative analysis for major complications by study era

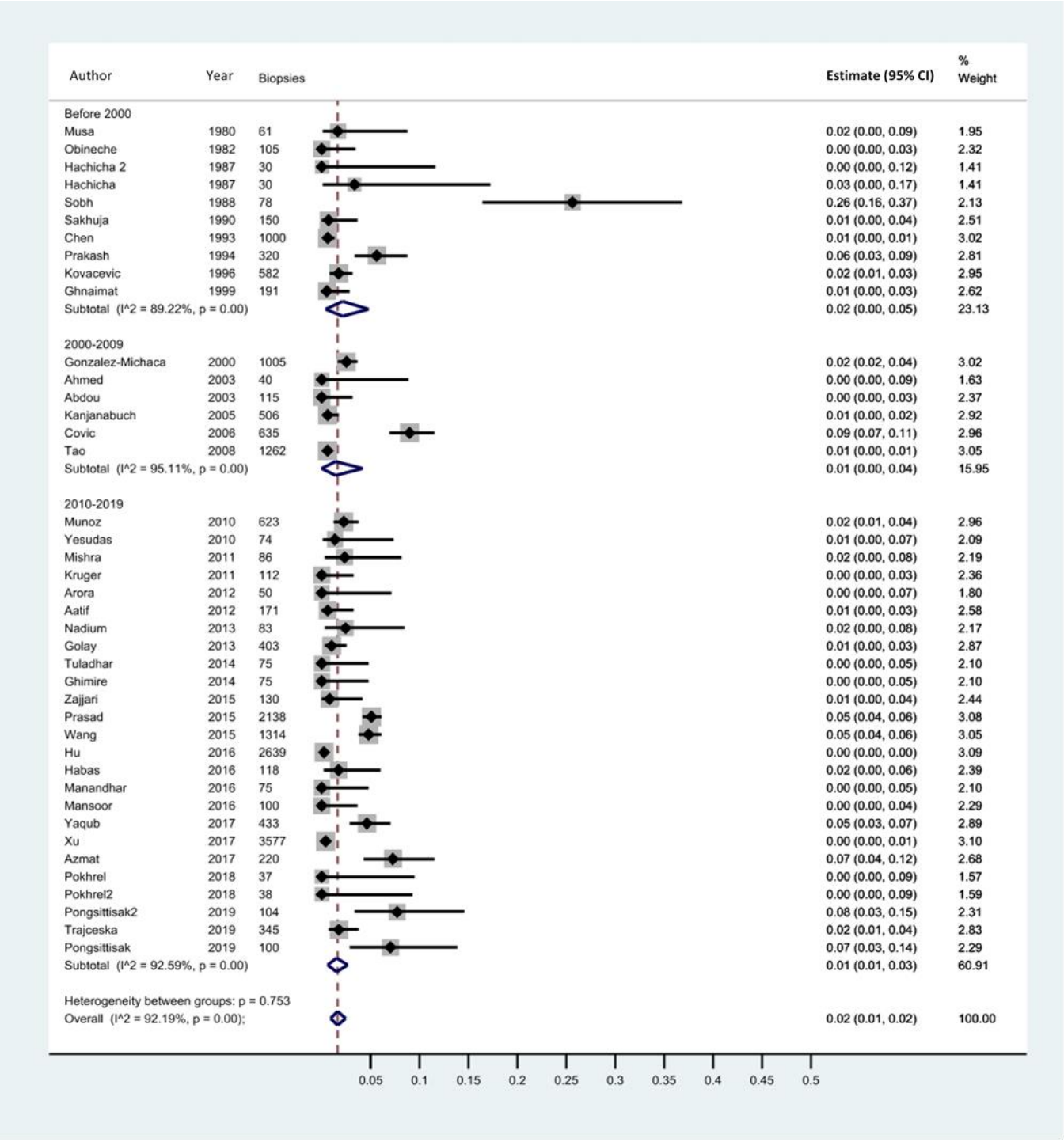


Figure S9: Major complications by region

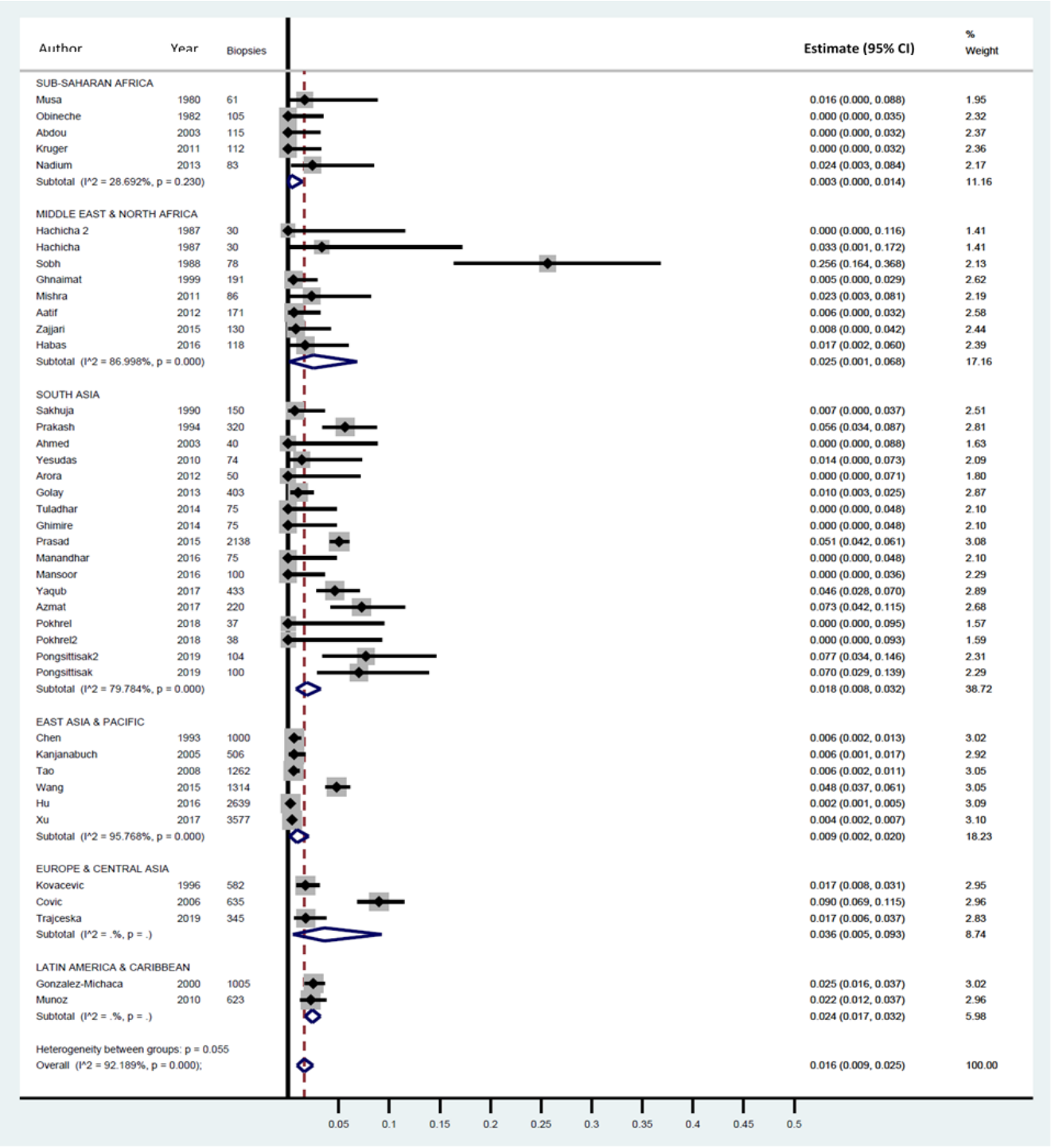


Figure S10: Major complications by income group

